A3-PS013-TC208

TraceLens: Incident Timeline and Evidence Collection for Small Teams

Threat Tracers
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Problem Description

Why small teams need a traceable and uncertainty-aware incident timeline

Incident evidence is often spread across host and application logs. A timeline is only useful when events can be traced to their sources and its limitations are made clear.

- Two required log formats may represent the same event differently.
- Clock offsets or unknown drift can make event ordering uncertain.
- Evidence may change after collection, while findings still depend on it.
- Manual correlation and triage can be slow and hard to reproduce.

PROJECT OBJECTIVE Propose a lightweight workflow that preserves original logs, builds a source-linked searchable timeline, surfaces uncertainty and integrity changes, and evaluates triage against a manual baseline.

Background & Context

From raw records to a reviewable timeline

An incident rarely arrives as one clean story: evidence is split across systems, timestamps and formats. TraceLens keeps each transformation traceable to the collected source.

1 Collect
Original files + metadata

→

2 Parse
Two PS-required formats

→

3 Normalize
Raw time + UTC + tolerance

→

4 Correlate
Source-linked event timeline

→

5 Review
Findings, limits + export

WHY INVESTIGATION GETS HARD

FORMAT — fields and timestamp conventions differ.
CLOCK — UTC fixes known offsets, not unknown drift.
INTEGRITY — later file changes can affect old findings.

WORKED EXAMPLE • TWO LOGS, TWO CLOCKS

auth.log: 10:00:02 +02:00 → 08:00:02 UTC
app.log: 08:00:04 UTC
Assume each clock may be ±3 s: ranges overlap, so event order is uncertain.

DESIGN RULE: preserve raw time + normalized UTC + justified tolerance + source reference; never imply certainty that the logs cannot support.

Literature & RFC Review

Existing foundations and the specific design response in TraceLens

REFERENCE	ESTABLISHED CONTRIBUTION	TRACE LENS RESPONSE
NIST SP 800-92 [1]	Computer-security log management.	Document collection and processing steps.
NIST SP 800-86 [2]	Forensic techniques in incident response.	Keep source provenance and analysis history.
RFC 3227 [3]	Evidence collection and archiving guidance.	Record collection metadata; preserve originals.
RFC 3339 [4]	Internet timestamp representation.	Parse offsets; retain raw time and normalized UTC.
NIST FIPS 180-4 [5]	Standardized cryptographic hash algorithms.	Fingerprint files and detect later differences.
Plaso / Timesketch [6,7]	Existing log parsing and forensic timeline tools.	Evaluate linked uncertainty, review and impact alerts.

POSITIONING: established tools cover timelines [6,7]; cognitive-bias research [8] motivates staged review. TraceLens evaluates their connected use.

Proposed Solution

TraceLens links evidence collection, incident reconstruction and review

COLLECT Original logs → **PARSE** Two required formats → **NORMALIZE** UTC + uncertainty → **CORRELATE** Source-linked timeline → **REVIEW / EXPORT** Auditable findings

01 TIME UNCERTAINTY

Keep the raw time, normalize known offsets to UTC, and model justified clock tolerance as a range.

EXAMPLE • ASSUMED ±3 s CLOCK ERROR

Host A: 10:00:02 → 09:59:59–10:00:05
Host B: 10:00:04 → 10:00:01–10:00:07

Ranges overlap → order is uncertain

02 BLINDLENS

Record the first interpretation, reveal context, then save the revised view and the reason. "Unresolved" remains valid.

EXAMPLE • CONTEXT CHANGES REVIEW

Before: "Suspicious deletion"
Ticket revealed: "Scheduled cleanup plausible"

Record why; verify before concluding

03 MUTATION RADAR

Re-hash collected files against the protected baseline. Trace a mismatch to dependent events and findings.

EXAMPLE • ILLUSTRATIVE HASHES

Saved: 3F9A…8C21
Recheck: 91D2…02BF → mismatch

Events E12/E13 → review required

CORE PIPELINE: preserve • verify • two parsers • ledger • UTC • searchable viewer • export • benchmark

A warning triggers review; it does not prove malicious tampering, causation or guilt.

Solution Optimization & Evaluation

Compare the baseline workflow with the proposed review mechanisms

BASIC APPROACH	TRACE LENS CHANGE	MEASURE
Sort displayed times	Preserve raw time; normalize known offsets; flag overlapping intervals.	Conversion tests; ambiguous-order precision/recall.
Unlinked parsed rows	Attach evidence ID and source record to each event.	Source-traceability rate.
Manual hash checking	Verify baseline hashes; map changed files to dependent findings.	Change-detection and impact-map precision/recall.
Store only final view	Keep initial and revised BlindLens assessment with reason.	Review-record completeness; triage time vs manual baseline.

THREE CONTROLLED SYNTHETIC SCENARIOS

A NORMAL
Known events across log sources

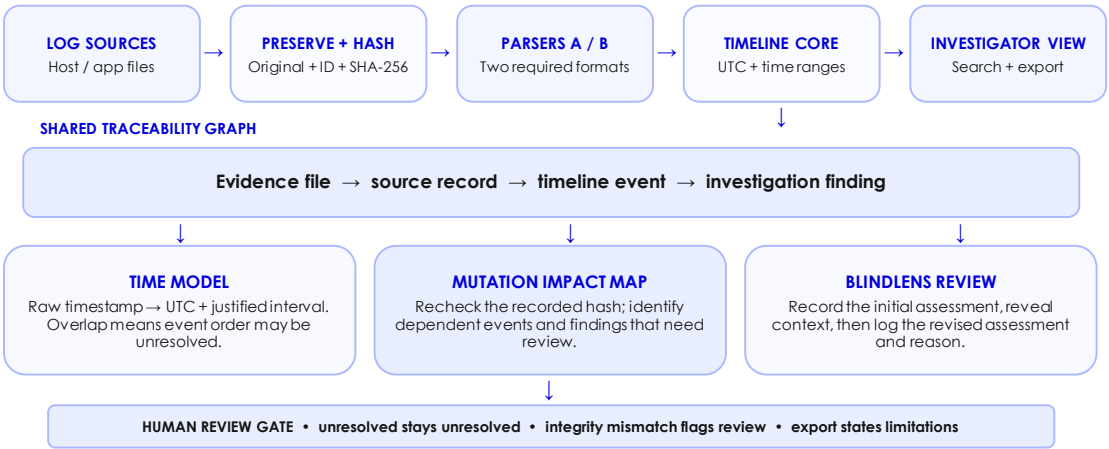
B CLOCK SKEW
Uncertain / overlapping time ranges

C CHANGED FILE
File modified after hash baseline

Report measured parsing accuracy, source traceability, hash detection, ambiguity handling and median triage time. Results are pending implementation.

Evidence Spine & Review Loop

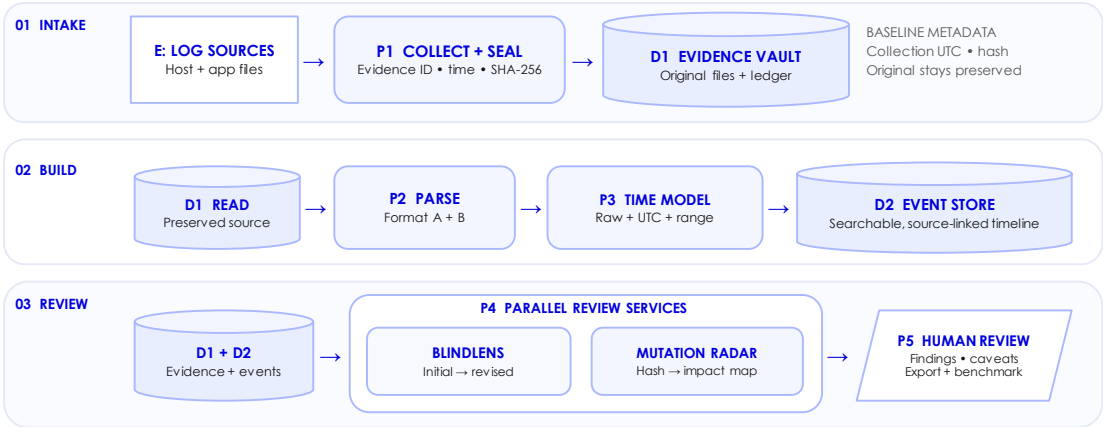
One preserved source of truth; three review mechanisms connect back to the same evidence.



Original files are not silently overwritten; every event and finding retains source links where available.

Data Flow Diagram

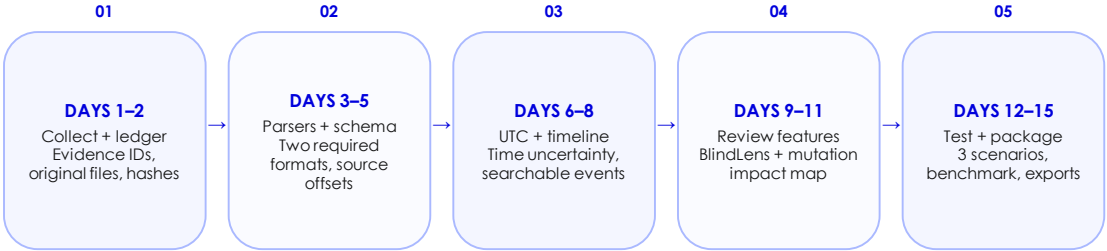
Evidence is preserved once; D1/D2 are referenced across processing and review stages.



KEY: rectangle = source • rounded block = process • cylinder = data store • arrows = data flow. Hash mismatch flags review; it does not prove intent.

Deployment Timeline

Build milestones are proposed and will be refined after mentor feedback.



LEVEL 1 • CURRENT SUBMISSION

Presentation PDF, team introduction video, and all repository setup steps.

LEVEL 2 • AFTER QUALIFICATION

Implement, test, submit code and provide the working demo as required by later steps.

Scope: local log evidence and transparent triage; no endpoint takeover, malware removal or forensic certification claims.

Bibliography

Selected standards, research and existing tools cited in this proposal

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Date and Time on the Internet: Timestamps (2002).

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[5] NIST FIPS 180-4

Secure Hash Standard (2015; update 1).

csrc.nist.gov/pubs/fips/180-4/upd1/final

[6] Plaso Documentation

Log2Timeline parser framework and super-timeline workflow.

plaso.readthedocs.io/en/latest/

[7] Timesketch

Open-source collaborative forensic timeline analysis.

timesketch.org

[8] Sunde & Dror (2019)

Cognitive and human factors in digital forensics: Problems, challenges, and the way forward. Digital Investigation, 29, 101–108.

doi.org/10.1016/j.diin.2019.03.011

Positioning note: existing tools cover much of the base workflow; TraceLens evaluates connected review mechanisms.

Thank You

Trace evidence. Show uncertainty. Make review accountable.

TraceLens is proposed as a transparent, source-linked incident timeline workflow for small teams.

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STAGE 1 / LEVEL 1 PROPOSAL • NO IMPLEMENTATION RESULTS CLAIMED